

PART A — MODULE 1 QUESTIONS

MATH2-M1-001

Prompt:

If $4(x - 2) = 20$, what is the value of x ?

- A) 5
- B) 7
- C) 12
- D) 18

MATH2-M1-002

Prompt:

At a market, 3 pounds of apples cost \$8.40. At this rate, what is the cost of 5 pounds of apples?

- A) \$11.20
- B) \$12.60
- C) \$14.00
- D) \$15.40

MATH2-M1-003

Prompt:

The daily high temperatures, in degrees Fahrenheit, for 5 days were 71, 69, 74, 67, and 70. What is the range of these temperatures?

- A) 4
- B) 5
- C) 7
- D) 9

MATH2-M1-004

Prompt:

A line has slope 3 and passes through the point (4, 17). Which equation represents the line?

- A) $y = 3x + 4$
- B) $y = 3x + 5$
- C) $y = 4x + 1$
- D) $y = 5x - 3$

MATH2-M1-005

Prompt:

A student has at most 60 minutes to exercise. The student spends 15 minutes warming up and then rides a bike for p minutes. Which inequality represents this situation?

- A) $p + 15 \geq 60$
- B) $p - 15 \leq 60$
- C) $p + 15 \leq 60$
- D) $15p \leq 60$

MATH2-M1-006

Prompt:

What is the solution to $x/3 + 5 = 11$?

- A) 2
- B) 6
- C) 16
- D) 18

MATH2-M1-007

Prompt:

The sum of two numbers p and q is 30. If $p = 2q$, what is the value of p ?

- A) 10
- B) 15
- C) 20
- D) 24

MATH2-M1-008

Prompt:

Which expression is equivalent to $7^5 \div 7^2$?

- A) 7^2
- B) 7^3
- C) 7^7
- D) 49^3

MATH2-M1-009

Prompt:

Which expression is equivalent to $4a - (a - 9)$?

- A) $3a - 9$
- B) $3a + 9$
- C) $5a - 9$
- D) $5a + 9$

MATH2-M1-010

Prompt:

One letter is chosen at random from the letters in the word MATHEMATICS. What is the probability that the letter chosen is M?

- A) $1/11$
- B) $2/11$
- C) $1/5$
- D) $2/9$

MATH2-M1-011

Prompt:

A triangle has a base of 14 centimeters and a height of 6 centimeters. What is the area, in square centimeters, of the triangle?

- A) 20
- B) 36
- C) 42
- D) 84

MATH2-M1-012

Prompt:

A circle has radius 5 inches. What is the area, in square inches, of the circle?

- A) 10π
- B) 25π
- C) 50π
- D) 100π

MATH2-M1-013

Prompt:

The equation $x(x - 8) = 0$ has two solutions. What is the greater solution?

- A) 0
- B) 4
- C) 8
- D) 16

MATH2-M1-014

Prompt:

If $f(x) = 7 - 4x$, what is $f(-3)$?

- A) -5
- B) 5
- C) 10
- D) 19

MATH2-M1-015

Prompt:

The number of students in a club increased from 50 to 58. By what percent did the number of students increase?

- A) 8%
- B) 12%
- C) 16%
- D) 20%

MATH2-M1-016

Prompt:

The table shows how 50 students travel to school.

Method: Bus, Train, Walk

Number of students: 24, 18, 8

What fraction of the students walk to school?

- A) $\frac{4}{25}$
- B) $\frac{8}{25}$
- C) $\frac{12}{25}$
- D) $\frac{21}{25}$

MATH2-M1-017

Prompt:

In a triangle, two angle measures are 48° and 67° . What is the measure of the third angle?

- A) 55°
- B) 65°
- C) 75°
- D) 115°

MATH2-M1-018

Prompt:

What is the value of $\sqrt[3]{64} - \sqrt{25}$?

- A) -1
- B) 1

- C) 3
- D) 9

MATH2-M1-019

Prompt:

A researcher records the number of hours of sunlight and the height of several plants of the same species. The data show that plants with more hours of sunlight generally have greater heights. Which statement best describes the association between hours of sunlight and plant height?

- A) There is a positive association.
- B) There is a negative association.
- C) There is no association.
- D) The association is impossible to determine from the description.

MATH2-M1-020

Prompt:

The equation $P = 10 + 2.5n$ models the total number of points P a player has after completing n levels of a game. What does 10 represent in this model?

- A) The number of points earned per level
- B) The number of levels completed
- C) The number of points the player has before completing any levels
- D) The total number of points after 10 levels

MATH2-M1-021

Prompt:

The volume of a cylinder is $V = \pi r^2 h$, where r is the radius and h is the height. What is the volume of a cylinder with radius 2 inches and height 9 inches?

- A) 18π
- B) 24π
- C) 36π
- D) 81π

MATH2-M1-022

Prompt:

A train travels 144 miles in 4 hours at a constant speed. What is the train's speed, in miles per hour?

- A) 24
- B) 36
- C) 48
- D) 58

PART B — MODULE 2 QUESTIONS

MATH2-M2-001

Prompt:

What is the solution to $5(2x - 1) = 3x + 16$?

- A) 2
- B) 3
- C) 5
- D) 7

MATH2-M2-002

Prompt:

The system of equations has solution (x, y) .

$$4x + 3y = 29$$

$$2x - 3y = 1$$

What is the value of $x - y$?

- A) 1
- B) 2
- C) 5
- D) 8

MATH2-M2-003

Prompt:

The graph of $y = (x - 1)(x - 6)$ crosses the x -axis at two points. What is the x -coordinate of the vertex of the graph?

- A) 2
- B) $\frac{5}{2}$
- C) $\frac{7}{2}$
- D) 6

MATH2-M2-004

Prompt:

For $x \neq 5$, which expression is equivalent to $(x^2 - 25)/(x - 5)$?

- A) $x - 5$
- B) $x + 5$
- C) $x^2 + 5$
- D) $x^2 - 5$

MATH2-M2-005

Prompt:

A sequence is defined by $a_1 = 6$ and $a_{n+1} = 2a_n + 1$ for each integer $n \geq 1$. What is the value of a_3 ?

- A) 13
- B) 19
- C) 27
- D) 31

MATH2-M2-006

Prompt:

A quantity of 200 is increased by 15% and then the result is decreased by 20%. What is the final quantity compared to the original quantity?

- A) 8% less
- B) 5% less
- C) 5% greater
- D) 8% greater

MATH2-M2-007

Prompt:

The table shows the number of books read by students in one month.

Books read: 0, 1, 2, 3

Number of students: 4, 7, 5, 4

What is the mean number of books read per student?

- A) 1.25
- B) 1.45
- C) 1.60
- D) 1.75

MATH2-M2-008

Prompt:

A trapezoid has bases of lengths 10 inches and 16 inches. The height of the trapezoid is 7 inches.

What is the area, in square inches, of the trapezoid?

- A) 63
- B) 81
- C) 91
- D) 112

MATH2-M2-009

Prompt:

In a right triangle, $\tan \theta = 5/12$. If the side adjacent to angle θ has length 36, what is the length of the side opposite angle θ ?

- A) 12
- B) 15
- C) 20
- D) 30

MATH2-M2-010

Prompt:

Line ℓ has equation $4x + 2y = 10$. A second line is perpendicular to line ℓ and passes through the point $(6, 1)$. What is the y-intercept of the second line?

- A) -4
- B) -2
- C) 2
- D) 4

MATH2-M2-011

Prompt:

The functions f and g are defined by $f(x) = x^2 - 2$ and $g(x) = 3x + 1$. What is $f(g(1))$?

- A) 8
- B) 10
- C) 14
- D) 18

MATH2-M2-012

Prompt:

What is the positive solution to $|2x - 5| = 9$?

- A) 2
- B) 5
- C) 7
- D) 9

MATH2-M2-013

Prompt:

A right triangle in the xy -plane has vertices $(0, 0)$, $(8, 0)$, and $(8, 6)$. What is the perimeter of the triangle?

- A) 20
- B) 22
- C) 24
- D) 28

MATH2-M2-014

Prompt:

For what value of k does the following system of equations have infinitely many solutions?

$$y = 3x + 2$$

$$6x - 2y = k$$

- A) -6
- B) -4
- C) 0
- D) 4

MATH2-M2-015

Prompt:

The table shows values of a function h .

x : $-2, -1, 0, 1$

$h(x)$: $0, 3, 4, 3$

Which equation could define h ?

- A) $h(x) = x^2 + 4$
- B) $h(x) = 4 - x^2$
- C) $h(x) = 2x + 4$
- D) $h(x) = 4 - 2x$

MATH2-M2-016

Prompt:

What is the solution to $x/(x + 4) = 2/3$?

- A) 4
- B) 6
- C) 8
- D) 12

MATH2-M2-017

Prompt:

A material has a density of 7 grams per cubic centimeter. What is the mass, in grams, of a sample of this material with volume 95 cubic centimeters?

- A) 600
- B) 665
- C) 720
- D) 780

MATH2-M2-018

Prompt:

Two similar polygons have corresponding side lengths of 5 centimeters and 8 centimeters. The perimeter of the smaller polygon is 30 centimeters. What is the perimeter, in centimeters, of the

larger polygon?

- A) 33
- B) 40
- C) 48
- D) 56

MATH2-M2-019

Prompt:

For a constant k , $(x + 4)(x + k) = x^2 + 11x + 28$ for all values of x . What is the value of k ?

- A) 4
- B) 7
- C) 11
- D) 28

MATH2-M2-020

Prompt:

One class has 12 students with a mean score of 84 on a quiz. Another class has 18 students with a mean score of 79 on the same quiz. What is the mean score for all 30 students?

- A) 80
- B) 81
- C) 82
- D) 83

MATH2-M2-021

Prompt:

For what value of a does the equation $a(x - 2) = 3x - 6$ have infinitely many solutions?

- A) 1
- B) 2
- C) 3
- D) 6

MATH2-M2-022

Prompt:

A rental company charges a fixed fee plus a constant amount per mile driven. A rental with 40 miles driven costs \$52, and a rental with 70 miles driven costs \$70. How many miles were driven for a rental that costs \$88?

- A) 80
- B) 90
- C) 100
- D) 110

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH2-M1-001 — Correct: 7 — Explanation: Divide both sides by 4 to get $x - 2 = 5$, then add 2 to get $x = 7$.

MATH2-M1-002 — Correct: \$14.00 — Explanation: The unit price is $\$8.40 \div 3 = \2.80 per pound, so 5 pounds cost $5 \times \$2.80 = \14.00 .

MATH2-M1-003 — Correct: 7 — Explanation: The range is the greatest value minus the least value: $74 - 67 = 7$.

MATH2-M1-004 — Correct: $y = 3x + 5$ — Explanation: A line with slope 3 has form $y = 3x + b$. Substituting (4, 17) gives $17 = 12 + b$, so $b = 5$.

MATH2-M1-005 — Correct: $p + 15 \leq 60$ — Explanation: The total time is p minutes plus 15 minutes, and this total is at most 60 minutes.

MATH2-M1-006 — Correct: 18 — Explanation: Subtract 5 from both sides to get $x/3 = 6$, then multiply by 3 to get $x = 18$.

MATH2-M1-007 — Correct: 20 — Explanation: Since $p = 2q$ and $p + q = 30$, then $2q + q = 30$, so $q = 10$ and $p = 20$.

MATH2-M1-008 — Correct: 7^3 — Explanation: When dividing powers with the same base, subtract exponents: $7^5 \div 7^2 = 7^3$.

MATH2-M1-009 — Correct: $3a + 9$ — Explanation: Distribute the subtraction: $4a - (a - 9) = 4a - a + 9 = 3a + 9$.

MATH2-M1-010 — Correct: $2/11$ — Explanation: MATHEMATICS has 11 letters, and 2 of them are M, so the probability is $2/11$.

MATH2-M1-011 — Correct: 42 — Explanation: The area of a triangle is $(1/2)bh$, so the area is $(1/2)(14)(6) = 42$.

MATH2-M1-012 — Correct: 25π — Explanation: The area of a circle is πr^2 . With $r = 5$, the area is $\pi(5^2) = 25\pi$.

MATH2-M1-013 — Correct: 8 — Explanation: Since $x(x - 8) = 0$, either $x = 0$ or $x - 8 = 0$. The greater solution is 8.

MATH2-M1-014 — Correct: 19 — Explanation: Substitute -3 for x : $f(-3) = 7 - 4(-3) = 7 + 12 = 19$.

MATH2-M1-015 — Correct: 16% — Explanation: The increase is $58 - 50 = 8$. The percent increase is $8/50 = 0.16$, or 16%.

MATH2-M1-016 — Correct: $4/25$ — Explanation: There are 50 students total and 8 walk, so the fraction is $8/50 = 4/25$.

MATH2-M1-017 — Correct: 65° — Explanation: The angle measures in a triangle sum to 180° . The third angle is $180 - 48 - 67 = 65^\circ$.

MATH2-M1-018 — Correct: -1 — Explanation: $\sqrt[3]{64} = 4$ and $\sqrt{25} = 5$, so $\sqrt[3]{64} - \sqrt{25} = 4 - 5 = -1$.

MATH2-M1-019 — Correct: There is a positive association. — Explanation: Since greater numbers of sunlight hours generally correspond to greater plant heights, the association is positive.

MATH2-M1-020 — Correct: The number of points the player has before completing any levels — Explanation: In $P = 10 + 2.5n$, the constant 10 is the value of P when $n = 0$.

MATH2-M1-021 — Correct: 36π — Explanation: Using $V = \pi r^2 h$ with $r = 2$ and $h = 9$ gives $V = \pi(2^2)(9) = 36\pi$.

MATH2-M1-022 — Correct: 36 — Explanation: Speed is distance divided by time, so $144 \div 4 = 36$ miles per hour.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH2-M2-001 — Correct: 3 — Explanation: Expand to get $10x - 5 = 3x + 16$. Then $7x = 21$, so $x = 3$.

MATH2-M2-002 — Correct: 2 — Explanation: Adding the equations gives $6x = 30$, so $x = 5$. Substituting gives $2(5) - 3y = 1$, so $y = 3$. Thus $x - y = 2$.

MATH2-M2-003 — Correct: $7/2$ — Explanation: The x -intercepts are 1 and 6. The vertex is halfway between them, so its x -coordinate is $(1 + 6)/2 = 7/2$.

MATH2-M2-004 — Correct: $x + 5$ — Explanation: Factor the numerator: $x^2 - 25 = (x - 5)(x + 5)$. For $x \neq 5$, the expression simplifies to $x + 5$.

MATH2-M2-005 — Correct: 27 — Explanation: $a_2 = 2(6) + 1 = 13$, and $a_3 = 2(13) + 1 = 27$.

MATH2-M2-006 — Correct: 8% less — Explanation: Increasing 200 by 15% gives 230. Decreasing 230 by 20% gives 184, which is 16 less than 200. Since $16/200 = 0.08$, the final quantity is 8% less.

MATH2-M2-007 — Correct: 1.45 — Explanation: The total number of books is $0(4) + 1(7) + 2(5)$

$+ 3(4) = 29$. There are 20 students, so the mean is $29/20 = 1.45$.

MATH2-M2-008 — Correct: 91 — Explanation: The area of a trapezoid is $(1/2)(b_1 + b_2)h$, so the area is $(1/2)(10 + 16)(7) = 91$.

MATH2-M2-009 — Correct: 15 — Explanation: $\tan \theta = \text{opposite/adjacent}$. Since $5/12 = \text{opposite}/36$, the opposite side length is $36 \times 5/12 = 15$.

MATH2-M2-010 — Correct: -2 — Explanation: The equation $4x + 2y = 10$ can be written as $y = -2x + 5$, so its slope is -2 . A perpendicular line has slope $1/2$. Through $(6, 1)$, the equation is $y - 1 = (1/2)(x - 6)$, which simplifies to $y = (1/2)x - 2$. The y-intercept is -2 .

MATH2-M2-011 — Correct: 14 — Explanation: First find $g(1) = 3(1) + 1 = 4$. Then $f(4) = 4^2 - 2 = 14$.

MATH2-M2-012 — Correct: 7 — Explanation: $|2x - 5| = 9$ gives $2x - 5 = 9$ or $2x - 5 = -9$. The solutions are $x = 7$ and $x = -2$, so the positive solution is 7.

MATH2-M2-013 — Correct: 24 — Explanation: The legs of the right triangle have lengths 8 and 6, so the hypotenuse is $\sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$. The perimeter is $8 + 6 + 10 = 24$.

MATH2-M2-014 — Correct: -4 — Explanation: The second equation can be rewritten as $y = 3x - k/2$. For infinitely many solutions, this must match $y = 3x + 2$, so $-k/2 = 2$ and $k = -4$.

MATH2-M2-015 — Correct: $h(x) = 4 - x^2$ — Explanation: Substituting $x = -2, -1, 0$, and 1 into $h(x) = 4 - x^2$ gives 0, 3, 4, and 3, matching the table.

MATH2-M2-016 — Correct: 8 — Explanation: Cross-multiply to get $3x = 2(x + 4)$, so $3x = 2x + 8$ and $x = 8$.

MATH2-M2-017 — Correct: 665 — Explanation: Density is mass divided by volume, so mass = density $\times$ volume. The mass is $7 \times 95 = 665$ grams.

MATH2-M2-018 — Correct: 48 — Explanation: The scale factor from the smaller polygon to the larger polygon is $8/5$. Perimeter scales the same way, so the larger perimeter is $30 \times 8/5 = 48$.

MATH2-M2-019 — Correct: 7 — Explanation: Expanding gives $(x + 4)(x + k) = x^2 + (k + 4)x + 4k$. Since this equals $x^2 + 11x + 28$, $k + 4 = 11$ and $k = 7$.

MATH2-M2-020 — Correct: 81 — Explanation: The combined mean is the total score divided by the total number of students: $(12 \times 84 + 18 \times 79)/30 = (1008 + 1422)/30 = 2430/30 = 81$.

MATH2-M2-021 — Correct: 3 — Explanation: For infinitely many solutions, the left side must be identical to the right side. Since $3x - 6 = 3(x - 2)$, a must be 3.

MATH2-M2-022 — Correct: 100 — Explanation: The cost increases by $\$70 - \$52 = \$18$ when the miles increase by 30, so the rate is $\$0.60$ per mile. The fixed fee is $\$52 - 0.60(40) = \28 . For a cost of $\$88$, solve $88 = 28 + 0.60m$, giving $m = 100$.